

AESG Berry Platform

AWS provisioning, cost estimate and go-live plan

Prepared for AESG by Strawberry Labs | 5 August 2026 | 250 named users

RECOMMENDED AWS / MONTH

\$485

DEFAULT AI ALLOWANCE / MONTH

\$5,000

INITIAL ROUTER INVOICE

\$3,157.50

Approve the Recommended AWS scenario for Berry Chat: one m7g.xlarge application host, dedicated Multi-AZ RDS PostgreSQL, S3, CloudWatch, ECR, Secrets Manager, KMS and backups. Keep inference, BerryCrawl and sandbox costs outside the AWS platform approval.

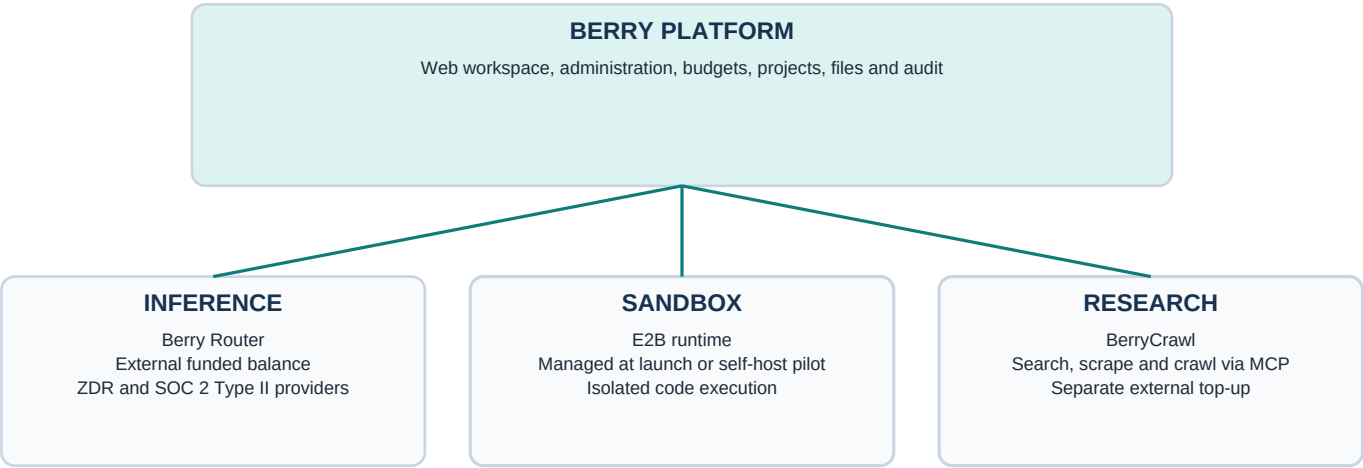
Executive position

- Berry platform hosting is modest for 250 named users: \$355.48/month minimum, \$485.35/month recommended, or \$681.69/month with two app nodes and an ALB.
- The recommended option spends the extra budget on Multi-AZ RDS. It deliberately avoids EKS, NAT Gateway, managed Redis and an application load balancer at launch.
- The single app host remains a documented failure point. It should be reproducible from a launch template, monitored and backed by a tested recovery runbook.
- Treat Google OIDC SSO, Gmail/Drive/Calendar connectors, DNS ownership, funding and restore/UAT evidence as P0 launch gates.
- Do not run self-hosted E2B on AWS c5.metal for this scale. The illustrative host alone is \$3,539.04/month before storage and operations.

Launch target: week commencing Monday, 10 August 2026. Proceed only after all P0 items are complete or explicitly waived by AESG's accountable owner.

1. Scope and platform model

This approval pack covers only the production Berry web platform and the AWS resources it needs. Cost Intelligence, proposal automation and the unified tool platform are excluded. Inference, research and sandbox execution are connected services with separate balances or infrastructure decisions.

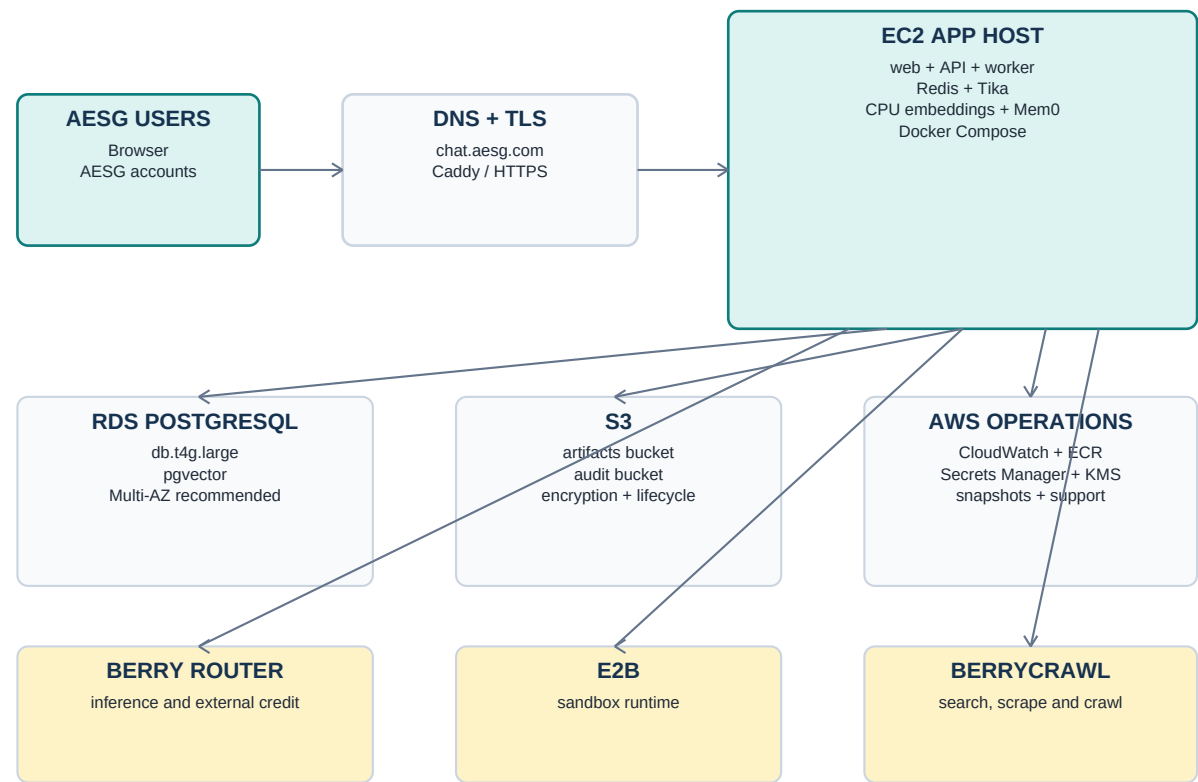


What Berry itself provides

Capability	Berry responsibility	External dependency
Workspace	Chat, projects, files, administration and audit	None for core UI
Budgets	Monthly user and department limits, including premium allocations	Berry Router still requires funded external credit
Inference	Provider routing configuration and usage controls	Berry Router and approved model providers
Research	MCP connection and user experience	BerryCrawl top-up
Sandbox	Job requests and results in the workspace	E2B managed or self-host runtime
Identity	Organization, roles, SSO configuration and SCIM endpoints	AESG Google Cloud and Workspace administration

Repository review: the production stack includes web, API, worker, PostgreSQL, Redis, Mem0, Tika, CPU embeddings and object storage. On AWS, S3 replaces MinIO and a dedicated Berry RDS can host separate Berry and Mem0 databases and roles.

2. Recommended AWS architecture



Cost-first design choices

Decision	Recommendation	Reason
Compute	1 x EC2 m7g.xlarge, Linux, 100 GB gp3	4 vCPU and 16 GiB avoid burst-credit risk for CPU embeddings
Database	RDS PostgreSQL db.t4g.large, 100 GB gp3, Multi-AZ	Dedicated Berry data plane with managed standby and backups
Object storage	Separate S3 artifact and audit buckets	Encryption, versioning, lifecycle and independent retention
Cache and support services	Redis, Tika, embeddings and Mem0 on app host	Avoid managed-service overhead at this scale
Edge	Caddy on EC2 with HTTPS	No ALB in recommended option; add it with second app node
Operations	CloudWatch, ECR, Secrets Manager, KMS and Systems Manager	Central visibility and no inbound SSH requirement
Containers	Docker Compose at launch	Kubernetes/EKS adds cost and operational complexity without a capacity need

Region assumption: eu-west-2 (London), matching the reference workbook. AESG must approve data residency. If me-central-1 or another region is required, re-run the AWS Calculator because prices and service availability will change.

3. AWS monthly cost estimate

MINIMUM	RECOMMENDED	HA APP
\$355.48	\$485.35	\$681.69

Scenario definitions

Scenario	Database	Application	Use
Minimum	Single-AZ RDS	1 app node, Caddy	Lowest cost; database AZ failure is accepted
Recommended	Multi-AZ RDS	1 app node, Caddy	Launch baseline; app node is replaceable but not HA
HA app	Multi-AZ RDS	2 app nodes plus ALB	Higher uptime after application state and deploy process are validated

Recommended monthly breakdown

Category	USD/month	Included
Compute and app storage	\$146.95	m7g.xlarge plus 100 GB EBS gp3
Network and DNS	\$18.05	public IPv4, transfer allowance and Route 53
Database	\$248.30	Multi-AZ RDS plus gp3 and backup allowance
S3 object storage	\$6.95	250 GB plus PUT/LIST/GET assumptions
Observability	\$11.98	20 GB logs, alarms and query/storage allowance
Registry and security	\$8.05	ECR, Secrets Manager and KMS
App snapshots	\$5.00	100 GB full-equivalent EBS snapshot
Business Support+	\$40.08	9% planning rate
Total	\$485.35	On-demand estimate before tax and commitments

Excluded: inference credits, BerryCrawl, E2B/sandbox, training, implementation labor, domain registration, taxes and third-party assessments. Evaluate Savings Plans or reservations only after 30-60 days of measured utilization.

4. Sizing, security and operating controls

The application is not expected to generate 250 simultaneous heavy requests. Most expensive work is external inference, research or sandbox execution. The AWS baseline should be monitored and scaled from measured latency, queue depth, CPU, memory and database behavior.

Required production controls

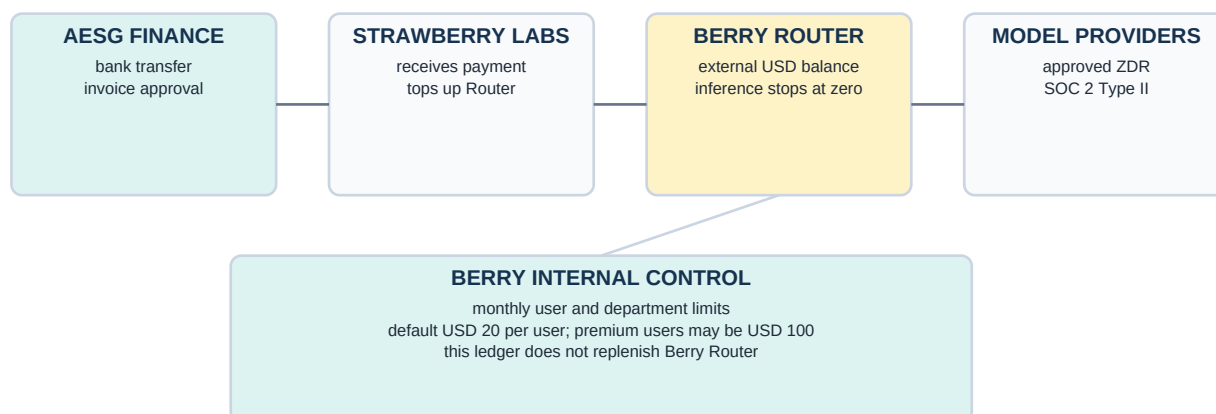
Area	Launch control	Evidence
Access	AESG-owned AWS account, root MFA, IAM Identity Center, least privilege	Account ID, roles and MFA record
Network	RDS private; expose only 80/443; administration through Systems Manager	Security-group review
Secrets	Secrets Manager and KMS; no credentials in repository, email or chat	Secret inventory and rotation owner
Storage	Block S3 public access; encryption, versioning and lifecycle	Bucket policy and KMS review
Backups	RDS automated backups and snapshots; EBS snapshots; restore drill	Measured restore time and application validation
Monitoring	Availability, 5xx, CPU, memory, disk, RDS and queue alarms	Alert reaches named on-call contact
Inference	Only approved ZDR and SOC 2 Type II providers	Provider allowlist and review owner
Audit	Immutable or tightly controlled audit bucket and retention policy	Retention approval and access log
Recovery	App launch template, image tag and documented redeploy	Replacement host drill

Scale triggers

- Add a second app node and ALB when app-host replacement time or availability requirements no longer fit the launch risk acceptance.
- Scale RDS vertically when sustained CPU, memory, IOPS or connection pressure is observed after query and pool tuning.
- Move Redis or embeddings to dedicated services only when measurements show resource contention or independent scaling is required.
- Keep BERRY_AUTH_MAX_USERS at least 250 and validate budget resets, audit durability and large-file behavior before launch.

The recommended option improves database availability but is not full application high availability. State that limitation explicitly in the approval.

5. Inference, research and commercial controls



Berry Router calculations

Funding case	Top-up	5% service charge	5% VAT on charge	Invoice
Suggested launch funding	\$3,000.00	\$150.00	\$7.50	\$3,157.50
Full default monthly allowance	\$5,000.00	\$250.00	\$12.50	\$5,262.50

Operating rules

- Berry resets internal user and department limits on the approved monthly cycle. Default is \$20 per user; approved users may receive \$100 or another limit.
- Berry Router is an external cash-like credit balance. Strawberry Labs replenishes it only after AESG payment and invoicing. If it reaches zero, inference stops even when Berry users still have internal allowance.
- Name the funding owner, approvers, low-balance threshold and replenishment response time before launch.
- BerryCrawl has a separate balance. The model uses a \$250 launch placeholder; AESG and Strawberry Labs must confirm the amount and low-balance alert.

Do not present Berry's \$5,000 monthly allowance as proof that Berry Router is funded. The two ledgers must be reconciled operationally.

6. Sandbox decision: E2B managed vs self-hosted

Recommendation: use managed E2B, or the already proven sandbox path, for launch. In parallel, run a single-node self-host pilot. Move to self-hosting only after load, isolation, failover, data-residency and operational ownership are proven.

Cost model

Option	Monthly cost	One-time	Capacity / limitation
Managed E2B at 2,500 runtime-hours	\$564.00	-	\$150 base plus \$0.1656 per active sandbox-hour
1 x Hetzner AX102-1 pilot	\$302.10	\$149.00	About 12 conservative active sandboxes; no HA
2 x Hetzner AX102-1	\$604.20	\$298.00	Hardware redundancy only; operations and edge unpriced
1 x AWS c5.metal	\$3,539.04	-	Host only; storage, E2B control plane and operations extra

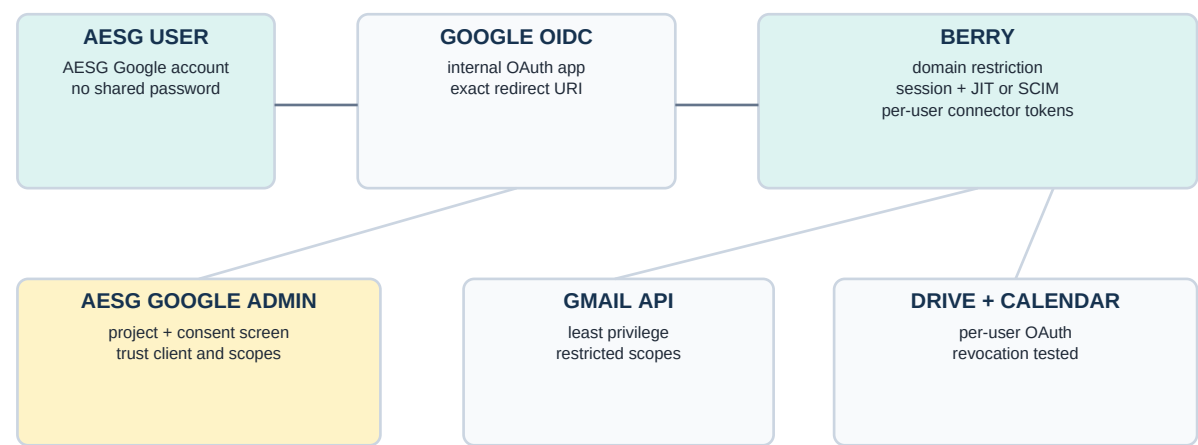
Runtime break-even before self-host operations

Comparison	Managed runtime-hours/month	Interpretation
Managed vs one AX102	about 918	Above this, pilot hardware is cheaper before staff, backup and edge
Managed vs two AX102 nodes	about 2,743	Above this, two-node hardware is cheaper before operations

Approval gates

- Firecracker needs KVM, bare metal or supported nested virtualization. AWS therefore pushes the design toward expensive bare-metal instances.
- E2B BYOC includes orchestrator, edge controller, monitoring, persistent storage and registry concerns. Orchestrator-node scaling is not fully automatic.
- Hetzner Germany/Finland requires AESG data-residency approval. Add VAT, IPv4, backups, traffic, edge and staff time before a purchase decision.
- The 12-concurrent-sandbox figure is a conservative planning assumption, not a guarantee. Load test AESG's actual template, CPU, RAM, disk and network behavior.

7. Google SSO, SCIM and connector readiness



P0 identity work

Work item	Strawberry Labs	AESG IT / Google admin	Acceptance
Google OIDC login	Implement and test login, domain restriction and session flow	Create AESG-owned internal OAuth app	AESG user signs in; external domain rejected
Credentials	Read client secret from AWS Secrets Manager	Provide client ID/secret securely; approve redirect URI	No secrets in source, email or chat
Lifecycle	Support JIT and SCIM path; test disabled users	Confirm Google provisioning capability and policy	Joiner, mover and leaver tests pass
Gmail	Implement least-privilege per-user OAuth connector	Trust approved restricted scopes	Read-only test and revocation pass
Drive	Implement search/read and shared-drive handling	Enable API and approve scopes	Search/read and revocation pass
Calendar	Implement read-only connector	Enable API and approve scopes	Calendar read and revocation pass

Repository review found SSO configuration surfaces and SCIM endpoints, but no verified end-to-end Google OIDC login flow and no verified Gmail, Drive or Calendar connector implementation. Treat these as code and configuration gates, not only credential collection.

Google administration notes

- Use an AESG-owned Google Cloud project with an internal consent screen and exact production redirect URI.
- Prefer per-user OAuth and the narrowest read-only scopes. Avoid domain-wide delegation unless AESG explicitly approves the security model.
- Gmail restricted scopes and Workspace API Controls can block the app until the AESG administrator trusts the client and scopes.
- If custom SCIM provisioning cannot be confirmed, an OIDC/JIT fallback requires explicit risk acceptance and a documented deprovisioning process.

8. Launch critical path

The dates below assume an accelerated approval and delivery window. They are not evidence that the underlying code, credentials, payments or infrastructure are complete.



No-go conditions

Gate	No-go if	Accountable decision
AWS	No AESG-owned account/card, root MFA, roles, region approval or resource inventory	AESG sponsor, finance and IT
DNS / TLS	chat.aesg.com owner or valid HTTPS endpoint is missing	AESG DNS owner
Identity	Google OIDC, domain restriction, lifecycle and admin trust are not tested	AESG identity admin
Connectors	Required Gmail/Drive/Calendar scope and revocation tests are incomplete	AESG sponsor and Google admin
Funding	Router or BerryCrawl balances and replenishment owners are not confirmed	AESG and Strawberry Labs finance
Reliability	Restore drill, monitoring, security review or UAT has not passed	AESG IT/security
Training	Revised scope/payment, PPTs or roster/session logistics are unresolved	AESG training coordinator
Sandbox	No approved managed launch path or safe pilot decision	AESG sponsor

Go/no-go review: Sunday, 9 August 2026. Every P0 item must be complete or explicitly waived with an owner, expiry date and rollback plan.

9. Go-live checklist - ownership, access and identity

Pri	Due	Task / decision	Accountable owner	Acceptance evidence
P0	Aug 5	Approve scope, region, RTO/RPO and single-app-host risk	AESG sponsor + IT security	Written approval and named decision owner
P0	Aug 5	Name AWS billing, DNS, Google admin, security and launch owners	AESG sponsor	Names and escalation path
P0	Aug 6	Create AESG-owned AWS account and attach corporate card	AESG finance + IT	Account ID and payment method validated
P0	Aug 6	Enable root MFA, IAM Identity Center, least privilege and budget alerts	AESG cloud admin	MFA, role list and alert test
P0	Aug 6	Provide DNS access for chat.aesg.com	AESG DNS owner	Change ticket and named operator
P0	Aug 8	Create DNS record and validate HTTPS certificate	AESG DNS + Strawberry Labs	Resolution and valid TLS chain
P0	Aug 7	Complete Google OIDC login implementation	Strawberry Labs engineering	AESG account login succeeds
P0	Aug 6	Create AESG Google Cloud project, internal consent and OAuth client	AESG Google admin	Project ID, client ID and internal audience
P0	Aug 7	Set exact redirect URI and store client secret in Secrets Manager	AESG Google admin + Strawberry Labs	Redirect test and secret reference
P0	Aug 8	Restrict AESG domain and test joiner/mover/leaver	AESG identity + Strawberry Labs	Allowed, blocked and disabled-user tests
P0	Aug 7	Confirm SCIM path or approve OIDC/JIT fallback	AESG identity admin	Provision/update/deprovision evidence
P0	Aug 8	Implement/test Gmail connector and least scopes	Strawberry Labs engineering	Connect, read and revoke test
P0	Aug 8	Implement/test Drive connector and shared drives	Strawberry Labs engineering	Search/read/revoke test
P1	Aug 9	Implement/test Calendar connector	Strawberry Labs engineering	Read-only and revoke test
P0	Aug 8	Trust OAuth client and scopes in Workspace API Controls	AESG Google admin	Admin configuration recorded

10. Go-live checklist - funding, infrastructure and assurance

Pri	Due	Task / decision	Accountable owner	Acceptance evidence
P0	Aug 7	Approve and clear \$3,000 Router top-up invoice	AESG + Strawberry Labs finance	\$3,157.50 received; balance visible
P0	Aug 7	Set Router funding owner, cadence, threshold and escalation	AESG + Strawberry Labs finance	Named approvers and response time
P0	Aug 8	Top up BerryCrawl and configure low-balance alert	AESG finance + Strawberry Labs	Research call succeeds; owner named
P0	Aug 8	Configure monthly Berry cycles, \$20 default cap and premium list	AESG program + Strawberry Labs	Admin export of limits and reset rule
P0	Aug 8	Provision EC2, RDS, S3, ECR, secrets, KMS, monitoring and backups	Strawberry Labs cloud engineer	Inventory and access review
P0	Aug 8	Deploy production images, migrations and max users >= 250	Strawberry Labs engineering	Health, version and migration recorded
P0	Aug 8	Create artifact and audit S3 buckets with required controls	Strawberry Labs cloud engineer	Policies, KMS and lifecycle reviewed
P0	Aug 9	Run RDS backup and restore drill	Strawberry Labs + AESG observer	Timestamped recovery evidence
P0	Aug 9	Configure service, capacity and external-balance alerts	Strawberry Labs cloud engineer	Test alert reaches on-call
P0	Aug 9	Review IAM, network, secrets, audit, providers and incidents	AESG security + Strawberry Labs	Signed checklist and exceptions
P0	Aug 9	Complete end-to-end UAT	AESG pilot + Strawberry Labs	Login, chat, files, research, budgets and audit pass
P1	Aug 9	Run representative load test and define scale thresholds	Strawberry Labs engineering	Latency/error report and headroom
P0	Aug 7	Approve managed sandbox launch path or self-host pilot	AESG sponsor + Strawberry Labs	Written decision and budget owner
P1	Aug 14	For self-hosting, approve residency and test isolation/failover	AESG security + Strawberry Labs	Load and security sign-off

11. Go-live checklist - training, launch and hypercare

Pri	Due	Task / decision	Accountable owner	Acceptance evidence
P0	Aug 5	Issue and approve revised training scope for 100 participants	Strawberry Labs + AESG sponsor	Signed scope, invoice and tax treatment
P0	Aug 7	Clear training deposit/payment	AESG finance	Payment receipt
P0	Aug 8	Share final PPTs, exercises and acceptable-use guidance	Strawberry Labs training lead	Approved facilitator materials
P0	Aug 7	Confirm 100-person roster, two cohorts and exact session times	AESG training coordinator	Roster and invitations
P0	Aug 9	Create support channel, on-call roster and severity rules	AESG IT + Strawberry Labs	Published runbook and contact test
P0	Aug 9	Hold go/no-go review	AESG sponsor + Strawberry Labs	P0 complete or documented waiver
P0	Aug 10	Controlled launch and first training session	AESG launch coordinator	Users sign in and complete guided tasks
P1	Aug 14	Daily service, spend and support review in first week	AESG IT + Strawberry Labs	Daily incident/spend log
P1	Aug 21	Post-launch review and 30-day optimization plan	AESG sponsor + Strawberry Labs	Approved cost/reliability/adoption actions

Training commercial working model

Item	Calculation	Amount
Working quote	100 people x AED 200	AED 20,000
Deposit	60% working assumption	AED 12,000
Final payment	40% balance	AED 8,000
Delivery	Two cohorts of 50; one 2-hour session per cohort	Week commencing Aug 10

This is a working quote before applicable VAT. The prior proposal covered 220 participants and a much broader 3-4 week program. Strawberry Labs should issue a revised scope and invoice before AESG makes payment.

12. Approval decisions and sources

Decisions required from AESG

Decision	Recommended answer	Named approver
AWS region	eu-west-2 for this estimate, or re-price approved alternative	TBD
Launch scenario	Recommended: one app host plus Multi-AZ RDS	TBD
App-host availability	Accept replaceable single host at launch	TBD
Sandbox	Managed launch path plus one-node self-host pilot	TBD
Google lifecycle	SCIM if confirmed; otherwise explicit OIDC/JIT fallback approval	TBD
Initial Router funding	\$3,000 principal; \$3,157.50 invoice	TBD
Training	Approve revised 100-person scope and payment	TBD

Key planning assumptions

- 250 named users; 730 service hours per month; on-demand AWS rates; USD cost model.
- 250 GB S3, 100 GB RDS, 100 GB app EBS, 20 GB logs and 250 GB outbound transfer per month.
- No AWS tax, implementation labor, commitment discount or sandbox operations labor is included.
- Managed E2B model assumes 2 vCPU, 4 GiB and \$0.1656 per active sandbox-hour plus \$150/month.
- Training VAT is not assumed until tax treatment is confirmed.

Primary sources

Source	URL
AWS EC2	https://pricing.us-east-1.amazonaws.com/offers/v1.0/aws/AmazonEC2/current/eu-west-2/index.json
AWS RDS	https://pricing.us-east-1.amazonaws.com/offers/v1.0/aws/AmazonRDS/current/eu-west-2/index.json
AWS S3	https://pricing.us-east-1.amazonaws.com/offers/v1.0/aws/AmazonS3/current/eu-west-2/index.json
AWS CloudWatch	https://pricing.us-east-1.amazonaws.com/offers/v1.0/aws/AmazonCloudWatch/current/eu-west-2/index.json
AWS Support	https://aws.amazon.com/premiumsupport/pricing/
E2B BYOC	https://e2b.dev/docs/byoc
E2B pricing	https://e2b.dev/pricing
Firecracker setup	https://github.com/firecracker-microvm/firecracker/blob/main/docs/dev-machine-setup.md
Hetzner price adjustment	https://docs.hetzner.com/general/infrastructure-and-availability/price-adjustment/
Google OIDC	https://developers.google.com/identity/openid-connect/reference
Google API Controls	https://support.google.com/a/answer/7281227
Gmail scopes	https://developers.google.com/workspace/gmail/api/auth/scopes

Detailed formulas, editable assumptions, complete line items and checklist tracking are in the accompanying Excel workbook.